1.4 Comparison of Decimals

Let us see Bob Beamon's Long Jump record of 1968 summer Olympics which stood for 23 years. His record was 8.90 *m*. This record was broken by Carl Lewis and Mike Powell in 1991,World championship held at Tokyo, Japan. Carl Lewis jumped 8.91 *m* and Powell jumped 8.95 *m*. Can you compare these distances?



We follow these steps to compare decimals.

1.4.1 Decimal Numbers with Equal Decimal Digits

Step:1 Compare the whole number part of the two numbers. The decimal number that has the greater whole number part is greater.

Step:2 If the whole number part is equal, then compare the digits at the tenths place. The decimal number that has the larger tenths digit is greater.

Step:3 If the whole number part and the digits at the tenths place are equal, compare the digits at the hundredths place. The decimal number that has the larger hundredth digit is greater. The same procedure can be extended to any number of decimal digits.

1.4.2 Decimal Numbers with Unequal Decimal Digits

Let us now compare the numbers 45.55 and 45.5. In this case, we first compare the whole number part. We see that the whole number part for both the numbers are equal. So, we now compare the tenths place. We find that for 45.55 and 45.5, the tenth place is also equal. Now we proceed to compare hundredth place. The hundredth place of 45.5 is 0 and that of 45.55 is 5. Comparing the hundredths place, we get $0 < 5$.

Therefore, $45.50 < 45.55$

Note

Zeros added to the right end of decimal digits do not change the value of that decimal number.

Example 1.13 Velan bought 8.36 *kg* of potato and Sekar bought 6.29 *kg* of potato. Which is heavier?

Solution

Compare 8.36 and 6.29

Comparing the whole number part, we get $8 > 6$.

Therefore, $8.36 > 6.29$

Example 1.14 Two automatic ice cream machines A and B designed to fill cups with 100 *ml* of ice cream were tested. Two cups of ice creams, one from machine A and one from machine B are weighed and found to be 99.56 *ml* from machine A and 99.65 *ml* from machine B. Can you say which machine gives more quantity of ice creams?

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